STEALTHRL: REINFORCEMENT LEARNING PARAPHRASE ATTACKS FOR MULTI-DETECTOR EVASION OF AI-TEXT DETECTORS

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ABSTRACT

AI-text detectors face a critical robustness challenge: adversarial paraphrasing attacks that preserve semantics while evading detection. We introduce *StealthRL*, a systematic threat model instantiation and robustness evaluation framework that stress-tests detector families under realistic adversarial conditions. StealthRL addresses the question: *how robust are detectors to black-box adaptive attacks trained explicitly to evade them?* Through reinforcement learning with multi-detector ensemble training, we demonstrate **catastrophic robustness failure**: detectors achieve near-zero true positive rates (0.001 mean TPR@1%FPR) under adaptive attack. Critically, attacks transfer across detector architectures, including a held-out detector family, revealing shared vulnerabilities rather than detector-specific brittleness. We also quantify the evasion-quality tradeoff across six attack settings (M0–M5), showing StealthRL matches the strongest evasion while preserving better judged quality than character-level obfuscation baselines. Our results expose fundamental limitations in current detection approaches and establish StealthRL as a principled adversarial evaluation protocol for robustness benchmarking. We release complete training and evaluation code for fully reproducible adversarial robustness assessment.

1 INTRODUCTION

AI-text detectors are deployed in academic integrity and content moderation, yet their robustness to adversarial attacks remains poorly understood. Standard benchmarks evaluate detectors on clean distributions, but adaptive attackers can iteratively refine paraphrases to evade detection. We study a realistic threat model: black-box adaptive attacks that query detector scores and optimize paraphrases to minimize detection confidence while preserving semantic content.

We present **StealthRL**, a reinforcement-learning framework that trains a paraphrase policy against detector ensembles to evaluate robustness under adversarial conditions. StealthRL addresses three questions: (1) Can attacks transfer across detector families? (2) Do detectors share common vulnerabilities? (3) What is the evasion-fidelity tradeoff? By evaluating at strict low-FPR operating points (1% false positive rate) and measuring transfer to held-out detectors, we provide systematic robustness evaluation that complements standard benchmarks.

Contributions.

- We implement black-box adaptive paraphrasing attacks via multi-detector RL training with semantic constraints, one of the first works to apply RL for adversarial detector evasion.
- We demonstrate strong evasion (0.001 mean TPR@1%FPR across three detector families) with cross-architecture transfer to a held-out detector.
- We establish an evaluation protocol measuring evasion, transfer, and fidelity at security-relevant operating points.
- We release complete training and evaluation scripts with full hyperparameters, enabling reproducible robustness benchmarking.

Anonymized code release. Code is available at `https://anonymous.4open.science/r/StealthRL-D42D`.

2 Related Work

Detector families for AI-text detection include curvature-based methods (DetectGPT and Fast-DetectGPT) and paired-LM detectors such as Binoculars. Mitchell et al. (2023); Bao et al. (2024); Hans et al. (2024)

Recent evasion work studies paraphrase attacks and RL-based humanization. AuthorMist demonstrates that reinforcement learning can train adaptive evasion policies against a single detector. David & Gervais (2025) Adversarial Paraphrasing studies universal attack strategies, while character-level attacks such as homoglyph substitution demonstrate strong evasion but often reduced readability. Cheng et al. (2025); Creo & Pudasaini (2025)

Parameter-efficient tuning methods such as LoRA and QLoRA enable efficient adaptation of large language models. Hu et al. (2021); Dettmers et al. (2023) Building on AuthorMist's RL framing, StealthRL extends to multi-detector ensemble training with held-out detector evaluation at strict low-FPR operating points, and provides complete open-source training and evaluation code for reproducible adversarial robustness assessment.

3 Method

Threat model. We assume black-box access to detector scores: the attacker can query detector confidence but does not require gradients. We evaluate transfer to a held-out detector (Binoculars) to test robustness beyond the training ensemble.

Training. Given AI-generated text x, we learn a paraphrase policy $\pi_\theta(y \mid x)$ that produces y with low detector confidence while preserving meaning. We optimize a composite reward:

$$R(x, y) = \alpha R_{\text{det}}(y) + \beta R_{\text{sem}}(x, y), \tag{1}$$

where R_{sem} is E5 embedding cosine similarity and $R_{\text{det}} = 1 - p(y)$ is the detector evasion reward, with $p(y)$ the AI probability. During training, $p(y) = 0.6 \cdot p_{\text{RoBERTa}}(y) + 0.4 \cdot p_{\text{Fast-DetectGPT}}(y)$ is the weighted average of two detectors. During evaluation, we test against all three detector families including the held-out Binoculars. We fine-tune the model using the Tinker API framework with GRPO Pang & Jin (2025) and LoRA adapters Hu et al. (2021) on Qwen3-4B-Instruct (rank 32, $\alpha = 32$, dropout 0.05) using group size 8, batch size 16, and three epochs over 10,000 MAGE train samples plus 200 dev samples. Reward weights: $\alpha = 1.0$, $\beta = 0.1$. Full hyperparameters in Appendix B.

4 Experimental Setup

We evaluate on MAGE test split (1,000 human + 1,000 AI samples, 100–500 tokens) Li et al. (2024) against three detector families: RoBERTa OpenAI, Fast-DetectGPT, and Binoculars (held-out during training). Mitchell et al. (2023); Bao et al. (2024); Hans et al. (2024) Baselines: M0 (no attack), M1 (simple paraphrase), M2 (StealthRL-v1, trained for 3 epochs using Tinker API), M3 (detector-guided selection), M4 (AuthorMist), and M5 (homoglyph substitution). David & Gervais (2025); Cheng et al. (2025); Creo & Pudasaini (2025) Metrics: TPR@1%FPR, ASR, AUROC, and E5 semantic similarity.

LLM-based quality evaluation. We additionally run an OpenAI `gpt-5-nano` Likert judge for paraphrase quality analysis. For fair cross-method comparison, we evaluate a shared subset of 200 AI samples per method (M1–M5), using identical sample IDs across methods. Each sample is scored on two 1–5 Likert axes: (i) overall linguistic quality (fluency/grammaticality/naturalness) and (ii) semantic similarity to the source.

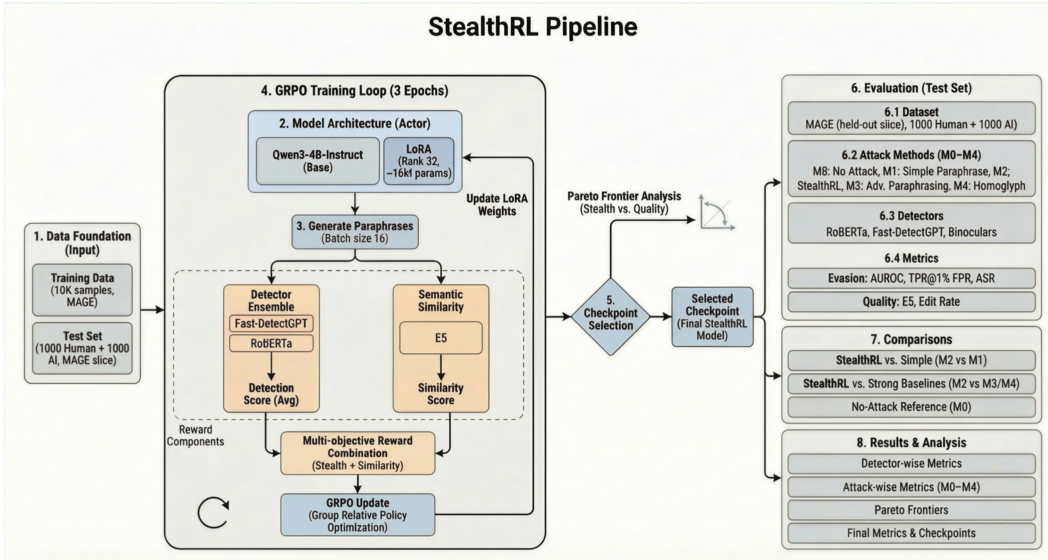

Figure 1: StealthRL training and evaluation pipeline. A single paraphrase policy is trained with detector-guided reward and quality constraints, then evaluated across multiple detector families with low-FPR metrics and tradeoff analysis.

Method	R (TPR@1%)	F (TPR@1%)	B (TPR@1%)	Mean ↓	ASR ↑	AUROC ↓
M0 No Attack	0.225	0.396	0.410	0.344	0.656	0.739
M1 Simple Para	0.095	0.103	0.039	0.079	0.921	0.589
M2 StealthRL-v1	**0.002**	**0.000**	0.001	**0.001**	**0.999**	**0.268**
M3 Adv. Para (guided)	0.104	0.089	0.045	0.079	0.921	0.595
M4 AuthorMist	0.085	0.015	0.010	0.037	0.963	0.454
M5 Homoglyph	**0.002**	**0.000**	**0.000**	**0.001**	**0.999**	0.437

Table 1: Main results on MAGE (TPR@1%FPR, ASR, and AUROC). Lower TPR/AUROC is better for the attacker; higher ASR is better. R/F/B denote RoBERTa/Fast-DetectGPT/Binoculars.

5 RESULTS

Table 1 summarizes detection evasion at 1% FPR across all methods M0–M5. StealthRL (M2) achieves near-zero mean TPR@1%FPR (0.001), reducing mean AUROC from 0.74 (no attack) to 0.27 with moderate semantic similarity (0.896 E5). Figure 2 visualizes comprehensive performance: panel (c) shows near-zero TPR across detectors at strict operating points, while panel (d) confirms 99.9% attack success rate. Qualitative examples of paraphrased outputs are provided in Appendix A.

Critically, strong transfer to the held-out Binoculars detector (0.001 TPR for M2) demonstrates that vulnerabilities are not detector-specific but reflect shared architectural weaknesses. Simple paraphrasing (M1) and detector-guided selection (M3) both remain at 0.079 mean TPR, while AuthorMist (M4) reduces mean TPR to 0.037. Homoglyph substitution (M5) and StealthRL (M2) both reach near-zero mean TPR, but StealthRL does so through learned paraphrasing rather than character-level obfuscation.

Likert quality results (gpt-5-nano). On the matched 200-sample subset per method, mean quality/similarity scores (1–5) are: M1 4.01/4.05, M2 2.59/2.67, M3 4.03/4.03, M4 3.64/3.70, and M5 2.01/2.94. These ratings quantify the quality-evasion tradeoff: M2 and M5 achieve strongest evasion, but M1/M3 preserve substantially higher judged readability and semantic faithfulness, with M4 in between.

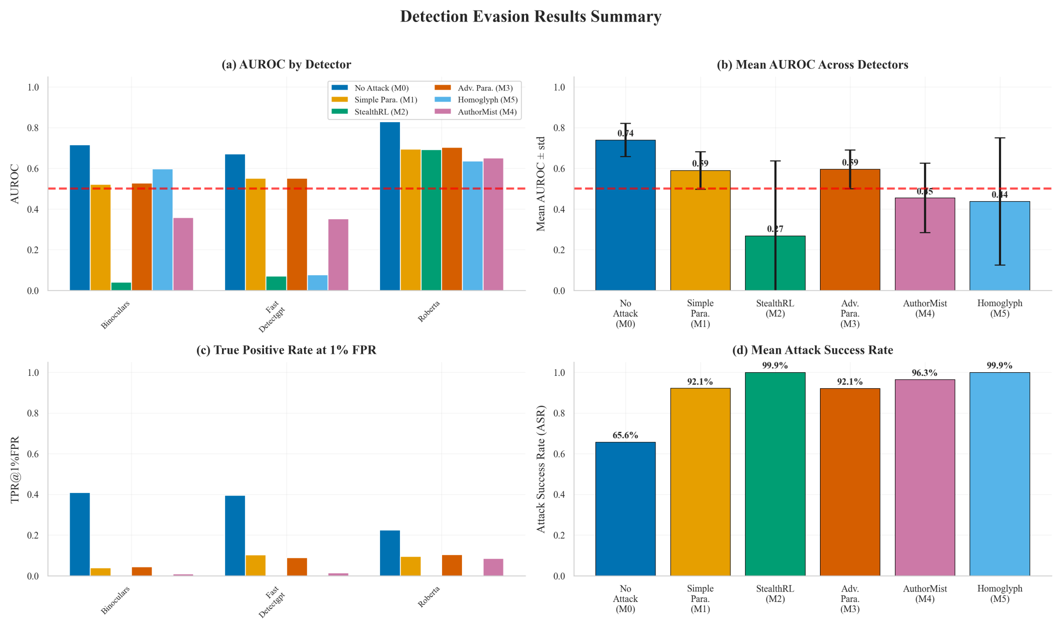

Figure 2: Detection evasion results summary for methods M0–M5. (a) AUROC by detector shows StealthRL-v1 (M2) dramatically reduces detector performance across all three detectors. (b) Mean AUROC demonstrates M2 achieves 0.27, indicating near-random classification. (c) TPR at 1% FPR shows near-zero detection rates for M2 at strict operating points. (d) Mean ASR confirms 99.9% attack success rate for StealthRL-v1, matching homoglyph-level evasion with better judged quality than M5 but lower quality than M1/M3.

Robustness implications. The catastrophic failure across detector architectures reveals fundamental vulnerabilities in current AI-text detection. Detectors rely on brittle statistical cues (token distributions, perplexity patterns, embedding geometry) rather than robust semantic understanding. Surface-level paraphrasing that preserves meaning suffices to evade detection, suggesting detectors learn superficial correlates of AI text rather than deeper linguistic features.

This robustness gap has critical security implications: adversaries can train adaptive attacks against deployed detectors, rendering them ineffective. The strong transfer to held-out architectures means ensemble defenses (combining multiple detectors) provide limited robustness improvement. Future work must develop semantic-aware detectors, adversarial training protocols, and provable robustness guarantees. Our evaluation framework provides a rigorous testbed for measuring progress on adversarially robust AI-text detection.

6 Limitations and Safety

Our evaluation focuses on three detector families (curvature-based, paired-LM, fine-tuned classifier) on one benchmark (MAGE). Broader coverage with watermark-based detectors, additional datasets, and multilingual evaluation remains important future work. We also do not explore defenses like adversarial training or certified robustness, which could improve detector resilience. Additionally, StealthRL achieves lower semantic fidelity (E5 similarity 0.896) compared to simpler baselines (M1: 0.960, M3: 0.976), indicating a quality-evasion tradeoff. Improving semantic preservation while maintaining strong evasion is an important direction for future work.

Safety and dual-use. Adversarial paraphrasing is dual-use technology. We position StealthRL as a *stress-testing and robustness evaluation tool* for researchers and detector developers, not a production evasion system. The near-zero TPR@1%FPR result exposes critical vulnerabilities that must be addressed before detectors are deployed in high-stakes applications. Our released code enables reproducible robustness evaluation and motivates defensive research into adversarially robust detection methods.

7 Conclusion and Future Work

StealthRL demonstrates catastrophic detector failure under adaptive attacks, revealing fundamental robustness gaps. Results motivate development of adversarially robust detection methods and establish rigorous evaluation protocols for AI-text detection security. Future work should explore adversarial training strategies, semantic-aware detectors, and provable robustness guarantees to defend against adaptive paraphrasing attacks.

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References

Guangsheng Bao, Yanbin Zhao, Zhiyang Teng, Linyi Yang, and Yue Zhang. Fast-detectgpt: Efficient zero-shot detection of machine-generated text via conditional probability curvature, 2024. URL `https://arxiv.org/abs/2310.05130`.

Yize Cheng, Vinu Sankar Sadasivan, Mehrdad Saberi, Shoumik Saha, and Soheil Feizi. Adversarial paraphrasing: A universal attack for humanizing ai-generated text, 2025. URL `https://arxiv.org/abs/2506.07001`.

Aldan Creo and Shushanta Pudasaini. Silverspeak: Evading ai-generated text detectors using homoglyphs, 2025. URL `https://arxiv.org/abs/2406.11239`.

Isaac David and Arthur Gervais. Authormist: Evading ai text detectors with reinforcement learning, 2025. URL `https://arxiv.org/abs/2503.08716`.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms, 2023. URL `https://arxiv.org/abs/2305.14314`.

Abhimanyu Hans, Avi Schwarzschild, Valeriia Cherepanova, Hamid Kazemi, Aniruddha Saha, Micah Goldblum, Jonas Geiping, and Tom Goldstein. Spotting llms with binoculars: Zero-shot detection of machine-generated text, 2024. URL `https://arxiv.org/abs/2401.12070`.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL `https://arxiv.org/abs/2106.09685`.

Yafu Li, Qintong Li, Leyang Cui, Wei Bi, Zhilin Wang, Longyue Wang, Linyi Yang, Shuming Shi, and Yue Zhang. Mage: Machine-generated text detection in the wild, 2024. URL `https://arxiv.org/abs/2305.13242`.

Eric Mitchell, Yoonho Lee, Alexander Khazatsky, Christopher D. Manning, and Chelsea Finn. Detectgpt: Zero-shot machine-generated text detection using probability curvature, 2023. URL `https://arxiv.org/abs/2301.11305`.

Lei Pang and Ruinan Jin. On the theory and practice of grpo: A trajectory-corrected approach with fast convergence, 2025. URL `https://arxiv.org/abs/2508.02833`.

A Qualitative Examples

Original (M0)	**StealthRL (M2)**
During cardio the heart increases its workload and all the body's other systems adjust to help support that endeavor. The blood vessels dilate, the muscles do their best to help pump blood back to the heart, and the lungs work harder to take in oxygen and remove waste gases like carbon dioxide.	In cardio, the heart ramps up its workload, prompting the body's other systems to adapt and assist. Blood vessels widen, muscles strive to push blood back to the heart, and lungs intensify their efforts to absorb oxygen and expel waste gases such as carbon dioxide.
The engine has to endure the torque of powering two axels and a drive shaft generally the transfer casing connects to the drive shaft with a ujoint and same for power distribution. The electricity goes into the battery first then is sent to the alternator where it generates voltage that powers all other electrical components like the lights, radio.	The engine must handle the torque from two axles and a drive shaft, typically linked via a universal joint in the transfer casing for power distribution. Electricity first charges the battery, then flows to the alternator, which converts it into voltage to power essential electrical systems such as lights and the radio.

Table 2: Representative paraphrases from the evaluation run (MAGE test split).

B Hyperparameters and Configuration

Parameter	Value
Model & LoRA	
Base model	Qwen/Qwen3-4B-Instruct-2507
LoRA rank	32
LoRA alpha	32
LoRA dropout	0.05
Training	
Algorithm	Group-Relative Policy Optimization (GRPO)
Learning rate	2.8×10^{-4}
Batch size	16
Group size	8
Epochs	3
Training samples	10,000 (MAGE train) + 200 (dev)
KL penalty coefficient	0.05
Reference policy	Qwen3-4B-Instruct (frozen)
Reward	
Detector weight (α)	1.0
Semantic weight (β)	0.1
Detector ensemble	RoBERTa (0.6) + Fast-DetectGPT (0.4)
Semantic metric	E5 embedding cosine similarity
Inference	
Temperature	1.0
Top-p	0.9
Max tokens	512
Prompt template	"Paraphrase the following text while preserving its meaning: [TEXT]"
Detectors	
RoBERTa OpenAI	openai-community/roberta-large-openai-detector
Fast-DetectGPT	Scoring model: EleutherAI/gpt-neo-2.7B
Binoculars	Lightweight: gpt2-medium + gpt2-large (held-out)
Evaluation	
Test samples	1,000 human + 1,000 AI (MAGE test)
Token window	100–500 tokens
FPR calibration	1% on 1,000 human samples (quantile)
Candidates per sample	1
Compute	
Training framework	Tinker API (Thinking Machines)
Reward computation	MacBook + NVIDIA A10 GPUs
Offline evaluation	MacBook + NVIDIA A10 GPUs
Seed	42

Table 3: Complete hyperparameters and configuration for reproducibility.